



大连海事大学
DALIAN MARITIME UNIVERSITY

图像/目标识别

辛文天

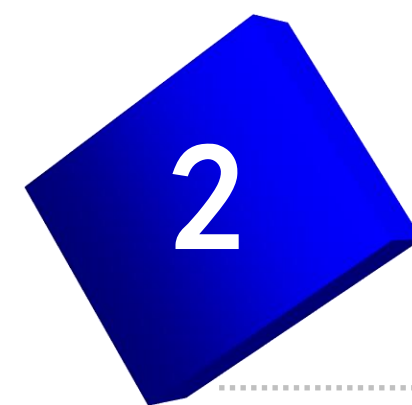
信息科学技术学院

2026/5/10

内容提纲



目标识别框架



Bag of Words

一、目标识别框架

表达 Representation

如何确定识别策略

- To handle intra-class variability, it is convenient to describe
使用概率模型来描述对象类别
an object categories using probabilistic models

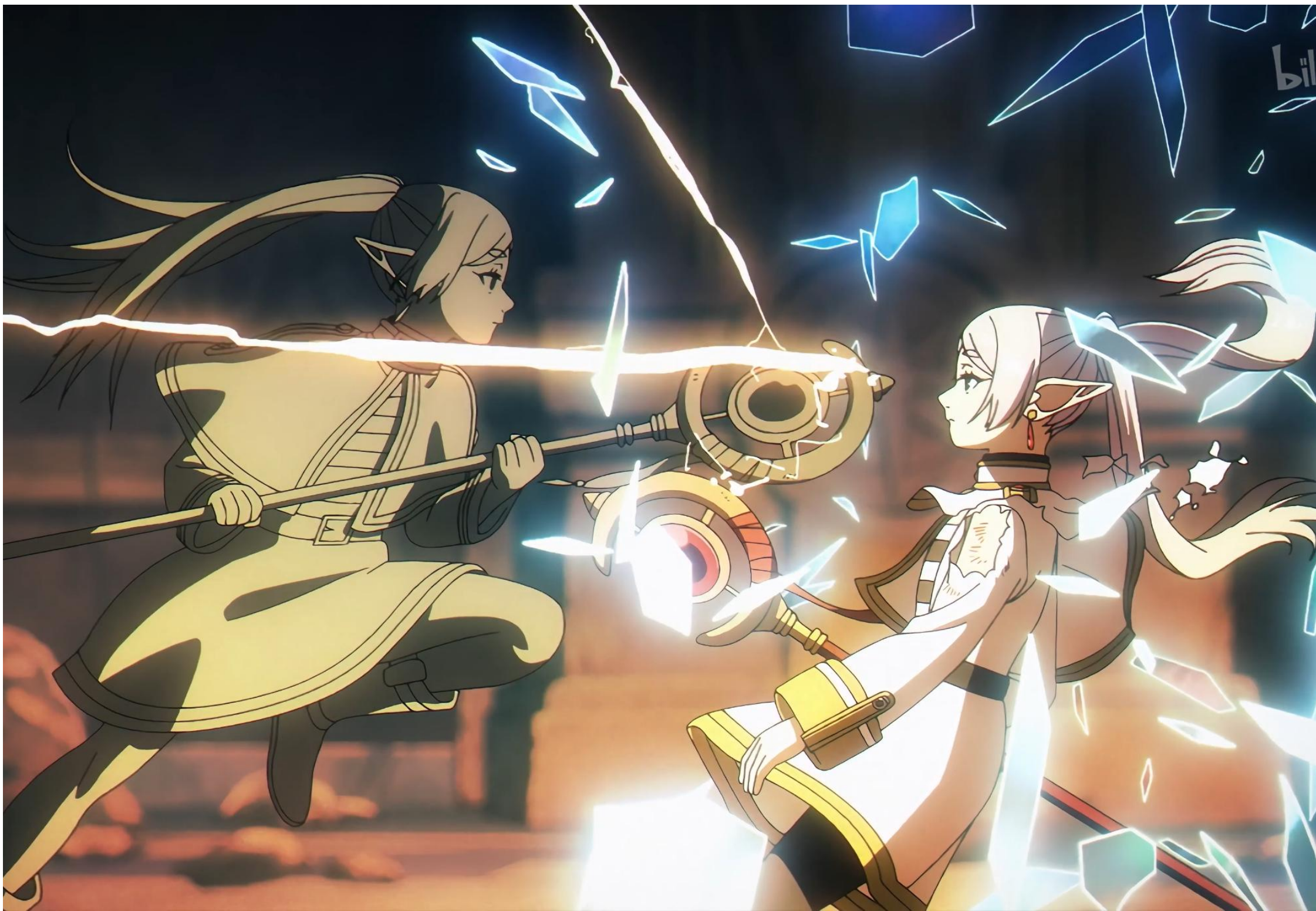
- Object models: Generative vs Discriminative vs hybrid

常用的识别方法有两个：一个是产生式一个是判别式

思考：模型的产生式和判别式的区别？

一、目标识别框架

表达 Representation



VS



产生式 VS 判别式

产生式，了解内部发生的事情的规律；判别式，了解二者之间的差异

一、目标识别框架

对象分类：统计学观点



问：判别模型 建模的什么概率？

$$p(\text{zebra} \mid \text{image})$$

vs. 给定一个图像，告诉我是不是斑马，属于后验概率

$$p(\text{no zebra} \mid \text{image})$$

似然：是根据提供的样本观测到的一些东西

- Bayes rule: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

如果我们知道这张照片里确实是斑马，那么这张照片看起来像斑马的可能性有多大。斑马有一些明显的特征，比如黑白条纹。所以，如果一张照片里有黑白条纹，那么它看起来就更像斑马。至于黑白条纹的特征就是我们能否学习到的关键。

$$\underbrace{\frac{p(\text{zebra} \mid \text{image})}{p(\text{no zebra} \mid \text{image})}}_{\text{posterior ratio}} = \underbrace{\frac{p(\text{image} \mid \text{zebra})}{p(\text{image} \mid \text{no zebra})}}_{\text{likelihood ratio}} \underbrace{\frac{p(\text{zebra})}{p(\text{no zebra})}}_{\text{prior ratio}}$$

先验：是预先得到的知识

一、目标识别框架

对象分类：统计学观点

- Discriminative methods model posterior
判别模型 建模的是后验
- Generative methods model likelihood and prior
产生模型 建模的是似然和先验

$$\underbrace{\frac{p(\text{zebra} | \text{image})}{p(\text{no zebra} | \text{image})}}_{\text{posterior ratio}} = \underbrace{\frac{p(\text{image} | \text{zebra})}{p(\text{image} | \text{no zebra})}}_{\text{likelihood ratio}} \underbrace{\frac{p(\text{zebra})}{p(\text{no zebra})}}_{\text{prior ratio}}$$

斑马在图片中的特征

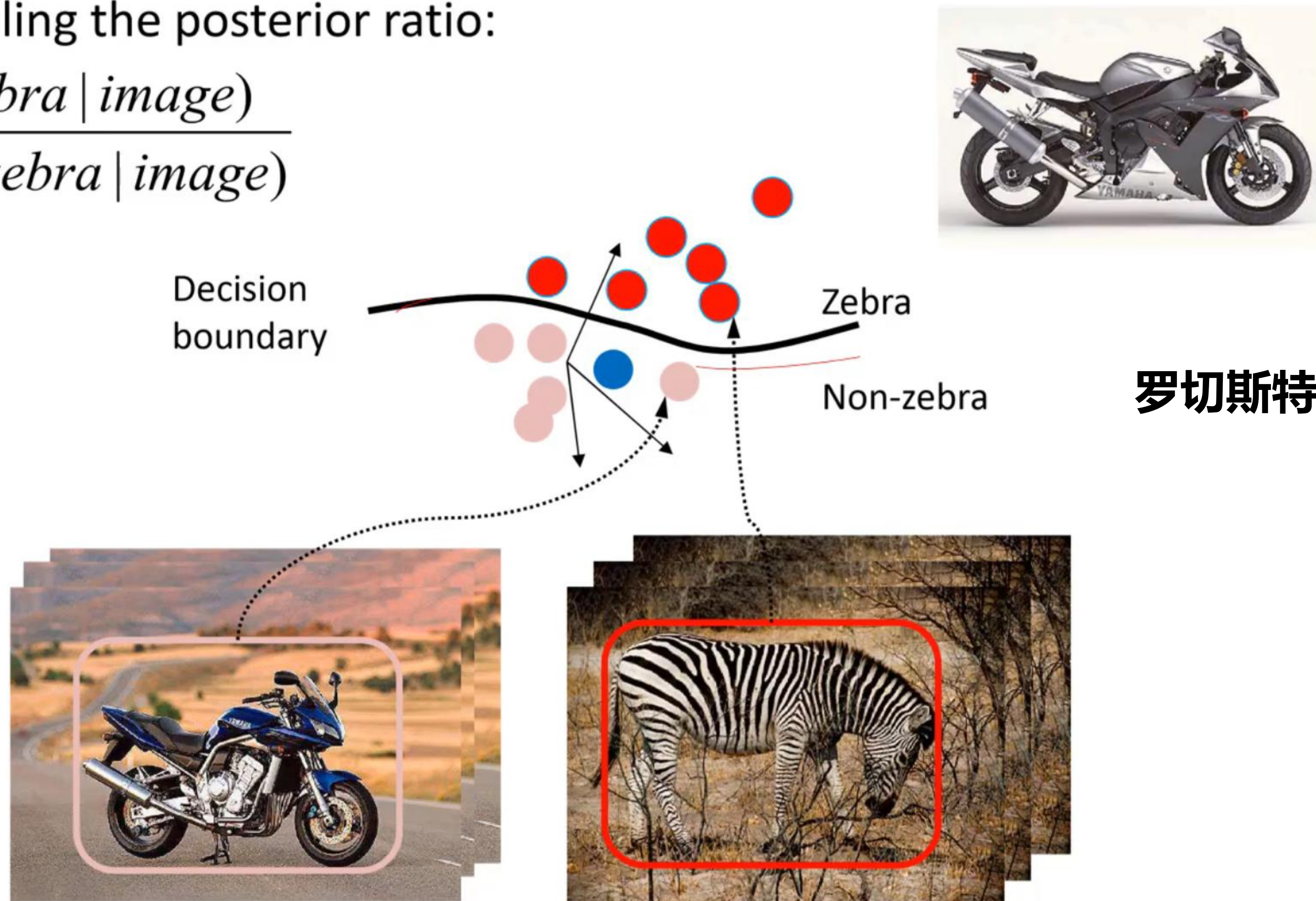
先验

一、目标识别框架

判别模型

- Modeling the posterior ratio:

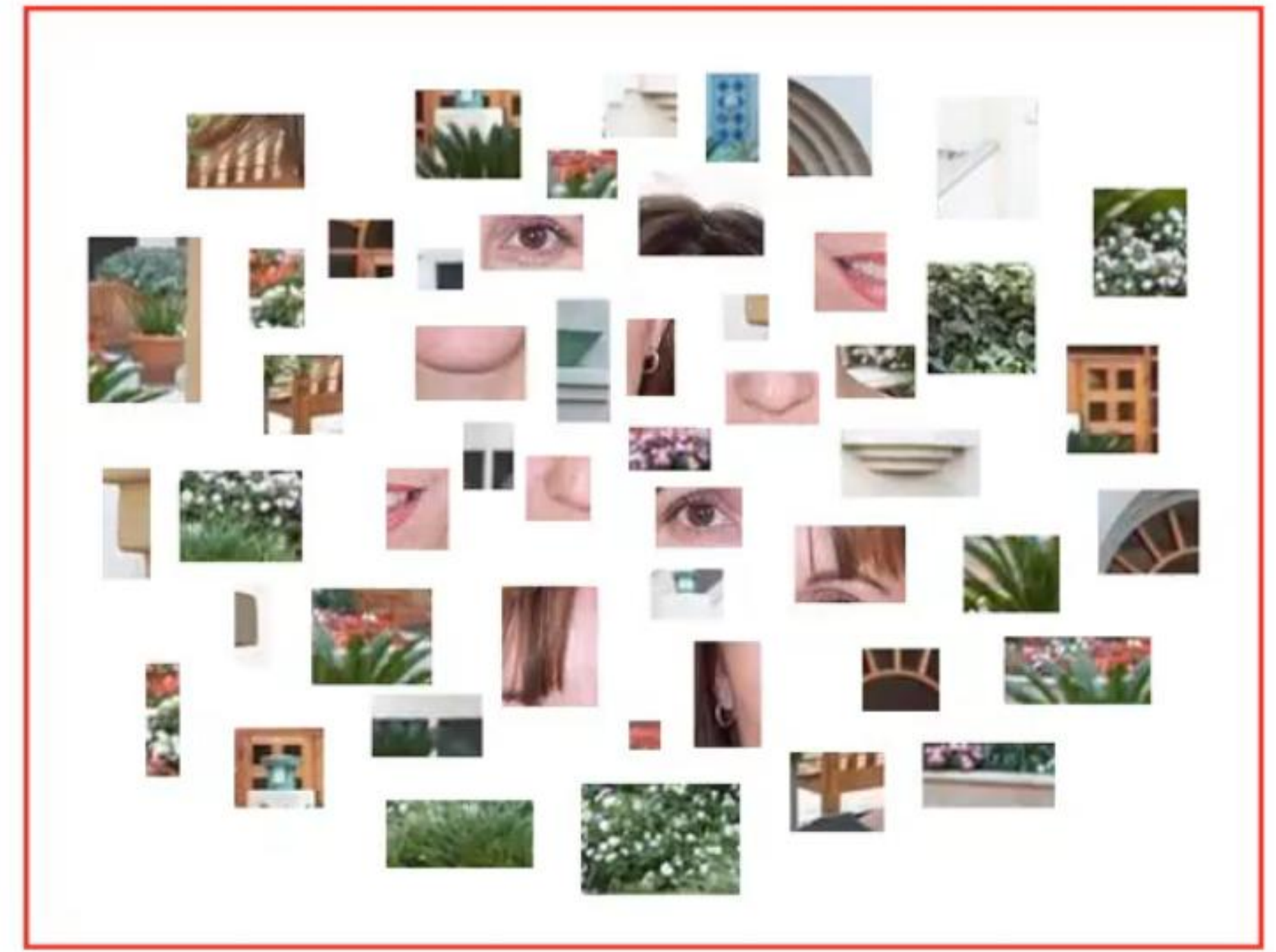
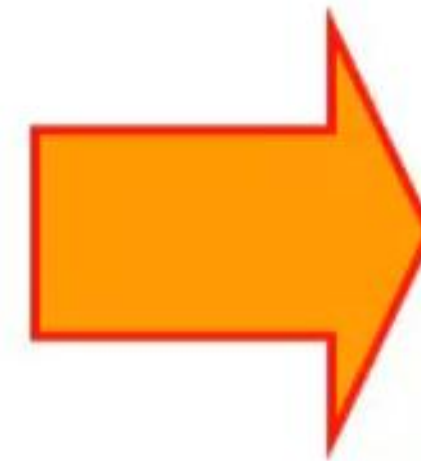
$$\frac{p(\text{zebra} | \text{image})}{p(\text{no zebra} | \text{image})}$$



罗切斯特回归

二、Bag of Words

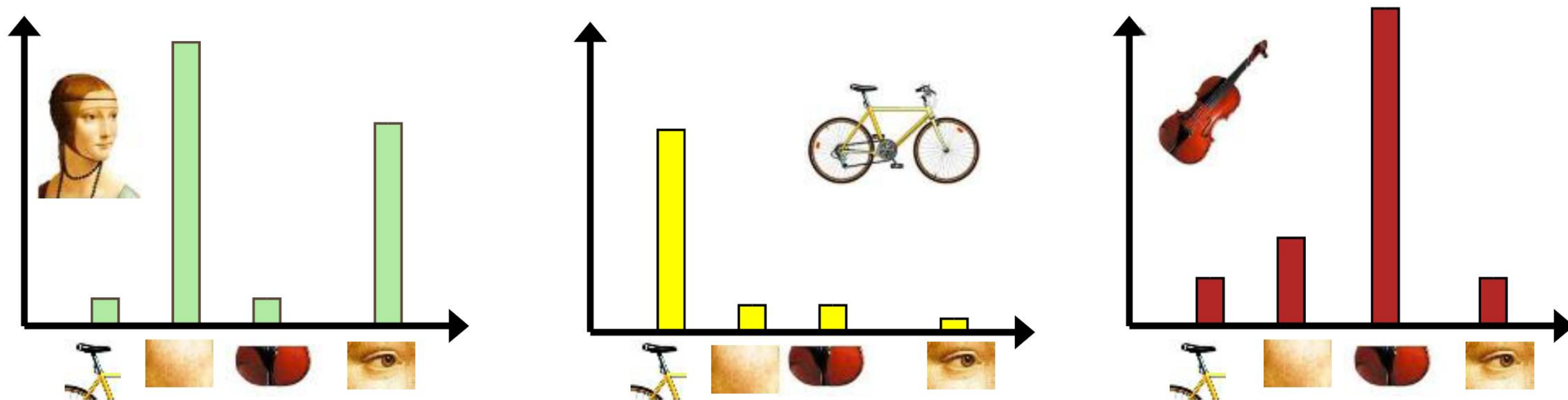
Bag of features models



二、Bag of Words

Bag of features steps

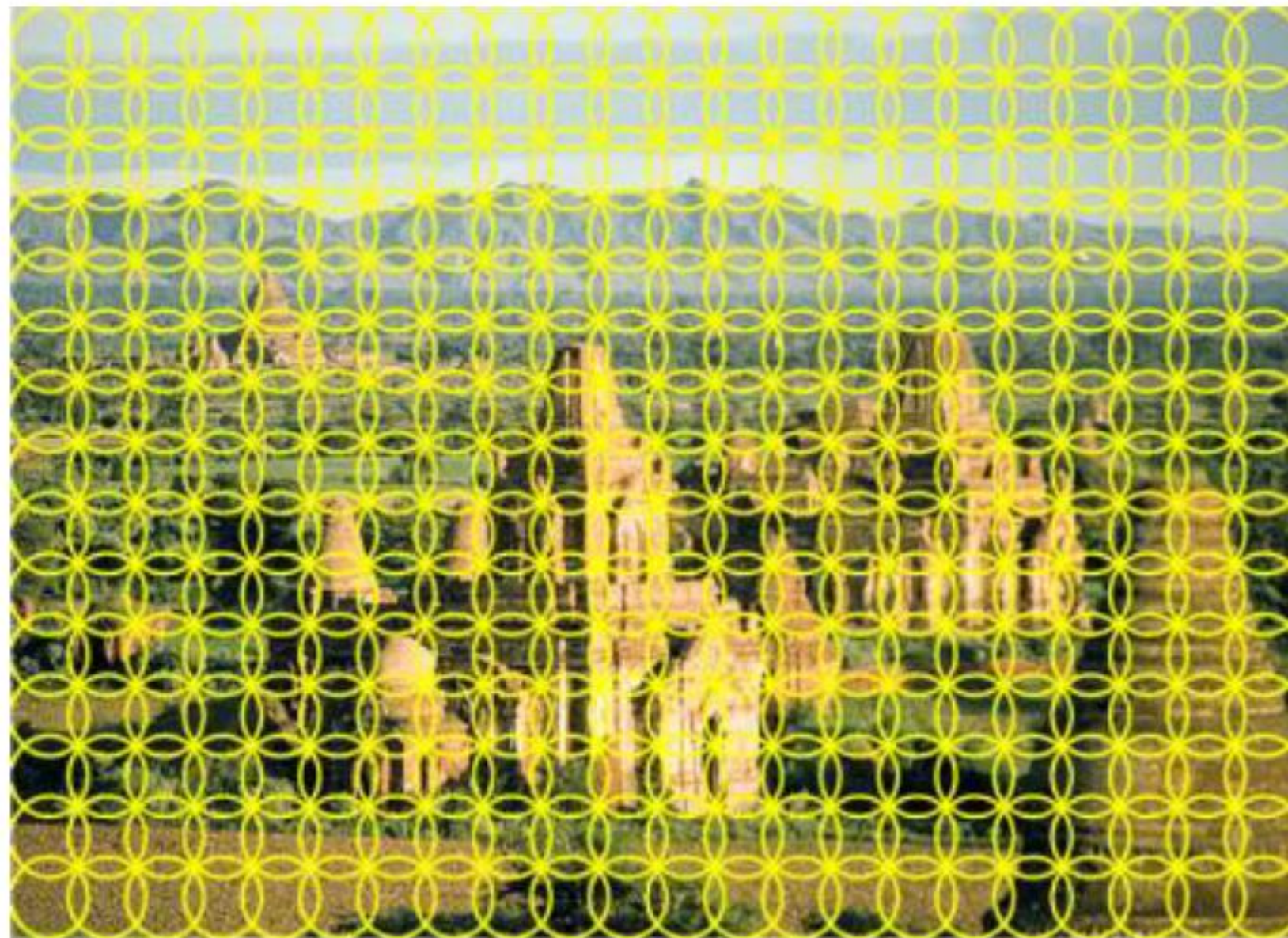
1. Extract features 提取特征
2. Learn “visual vocabulary” 学习视觉词汇表 (视觉词典)
3. Represent images by frequencies of “visual words” 把一张图片编码成 BoW 向量



二、Bag of Words

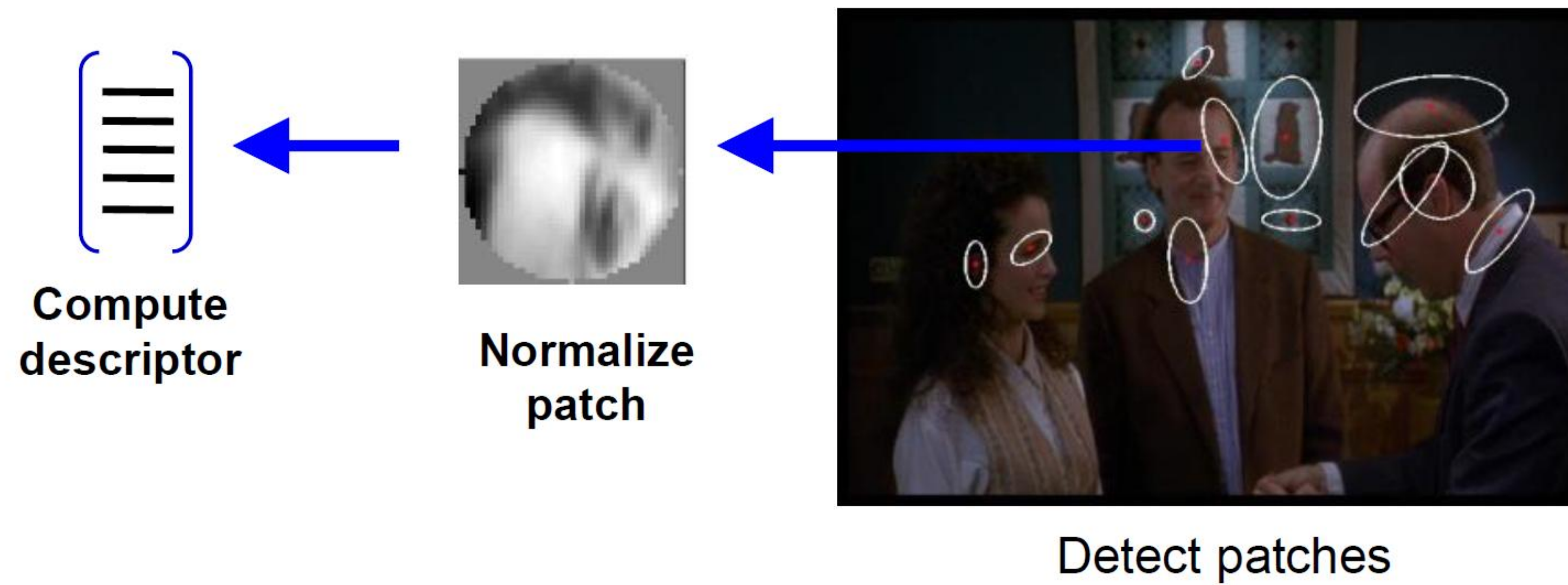
1. Feature Extraction

Regular grid or interest regions



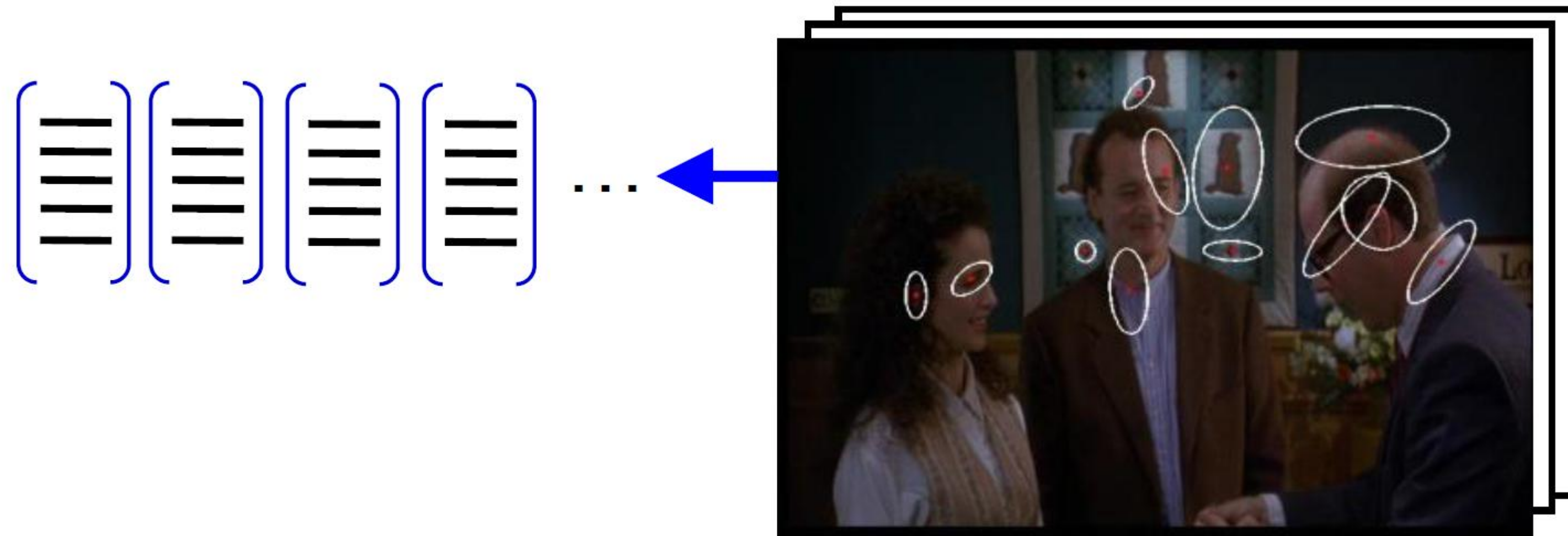
二、Bag of Words

1. Feature Extraction



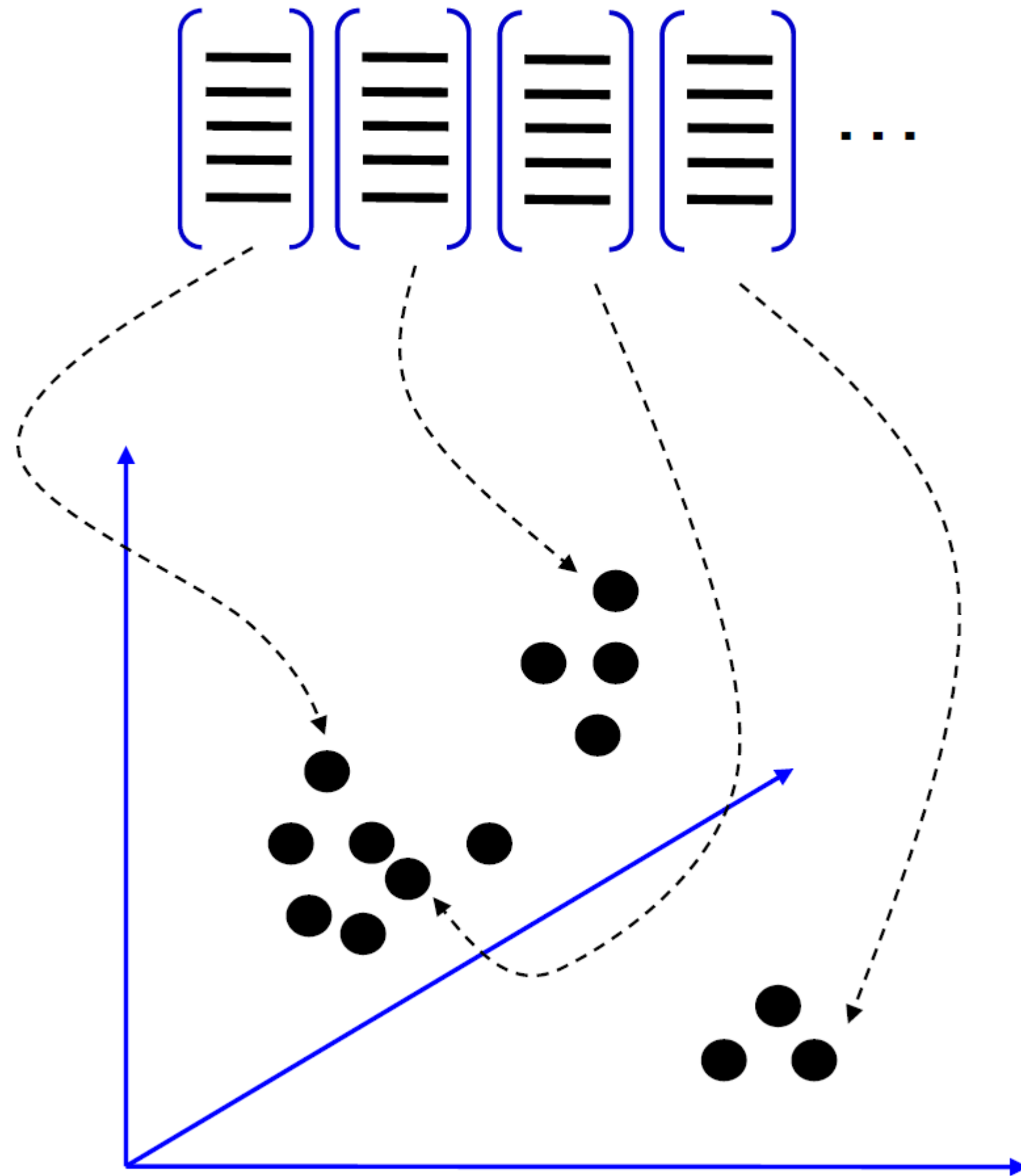
二、Bag of Words

1. Feature Extraction



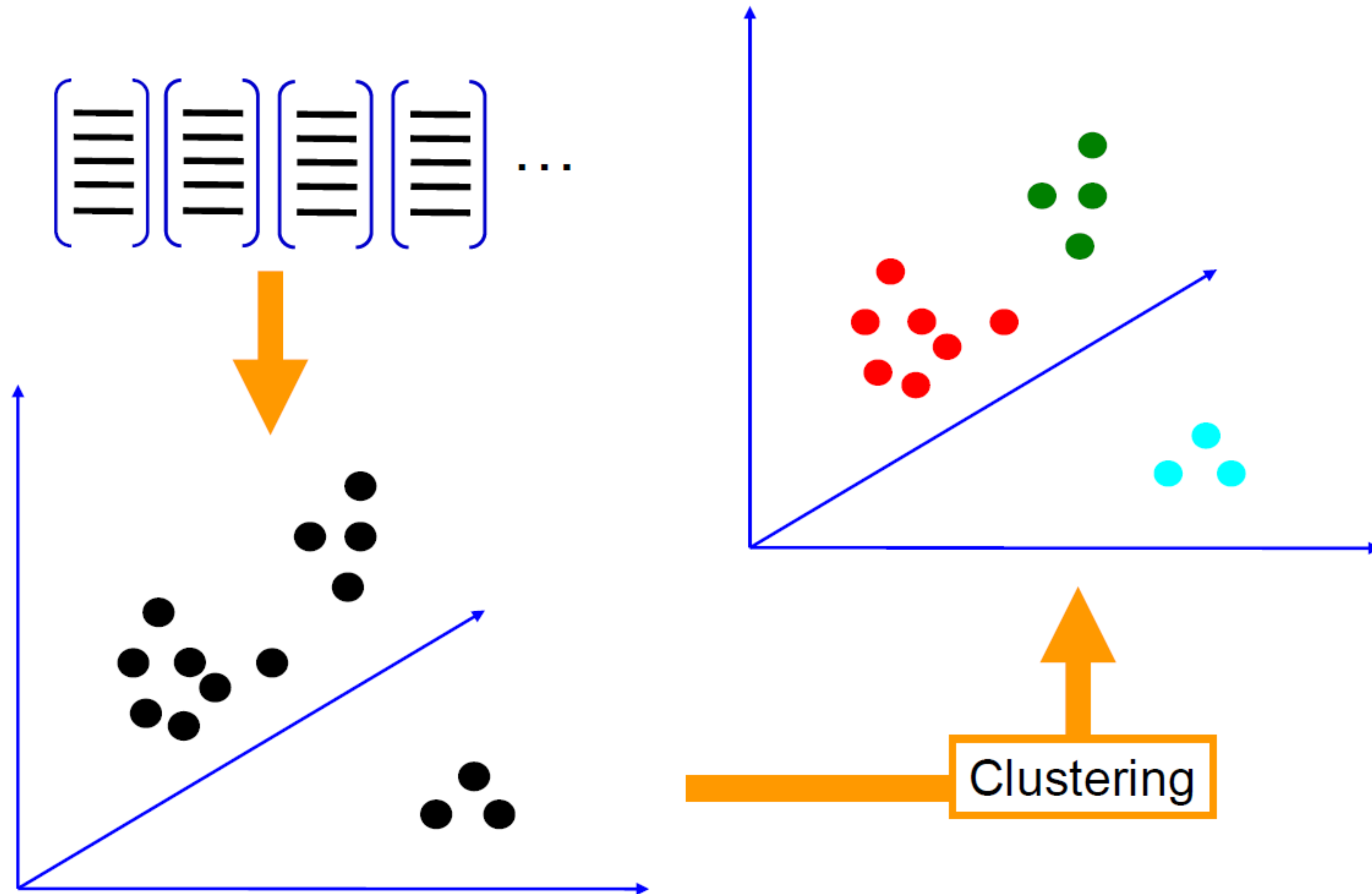
二、Bag of Words

2. Learning the visual vocabulary 学习视觉词汇表 (视觉词典)



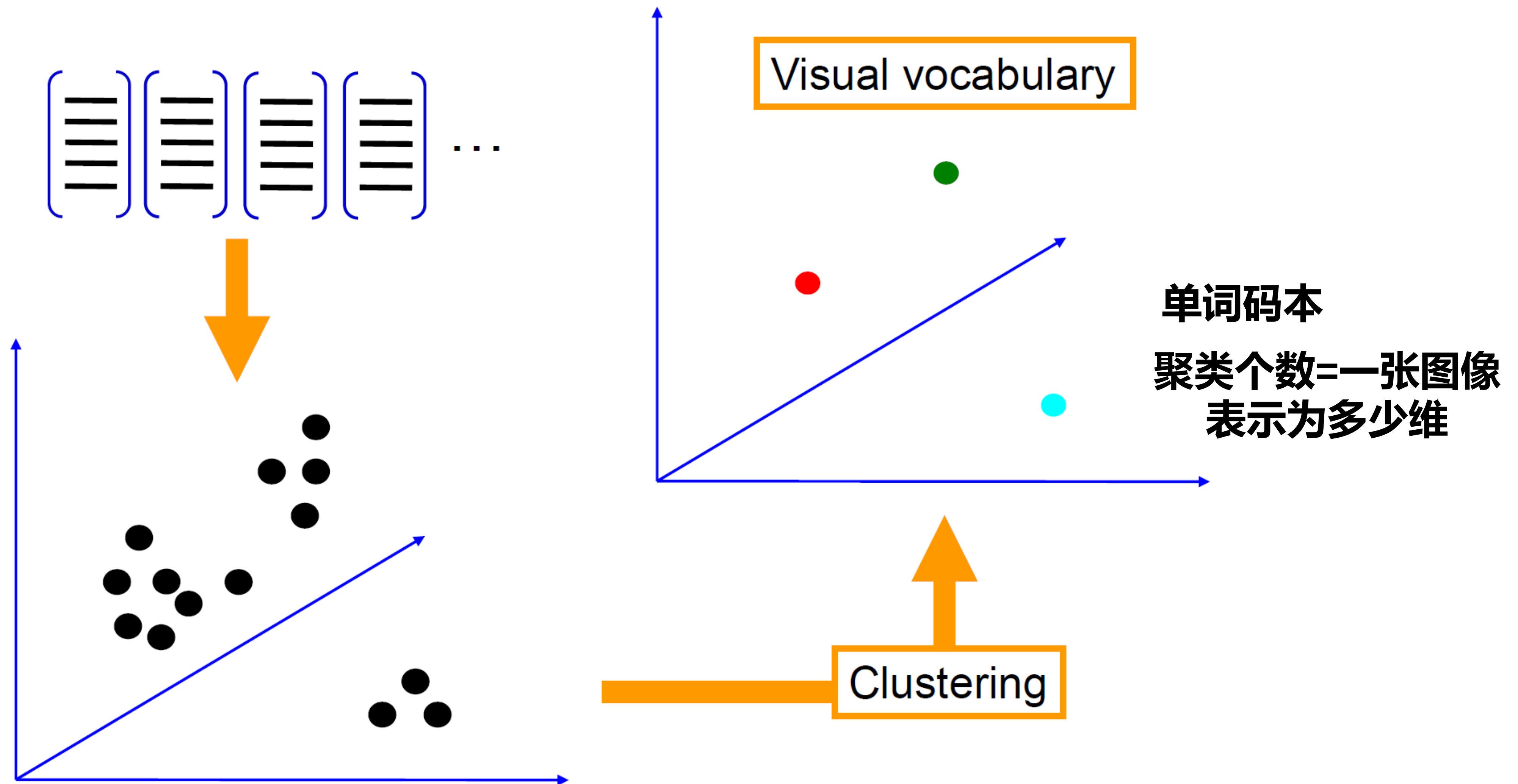
二、Bag of Words

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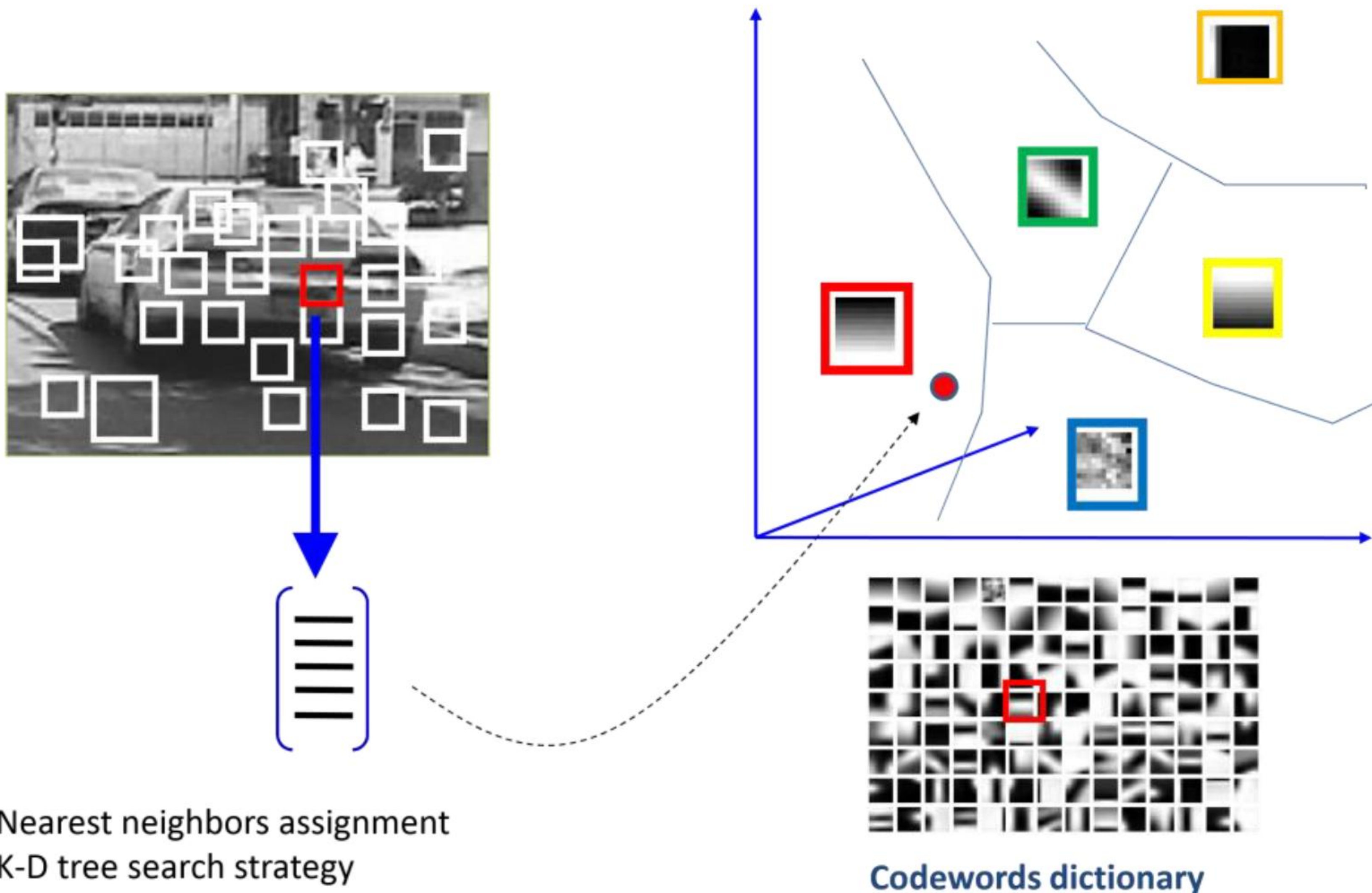
二、Bag of Words

2. Learning the visual vocabulary学习视觉词汇表（视觉词典）



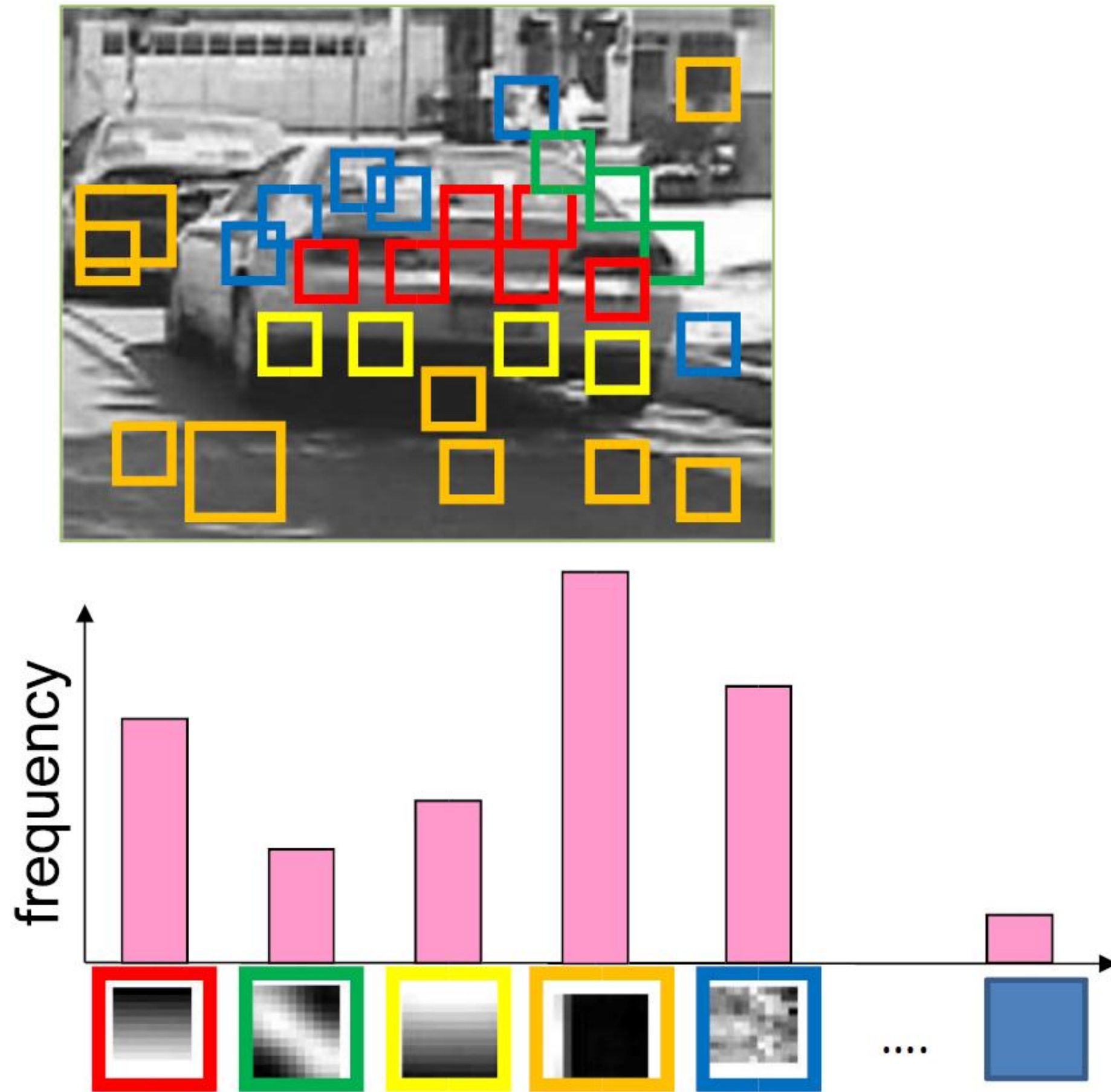
二、Bag of Words

3. Bag of Word representation 把一张图片编码成 BoW 向量

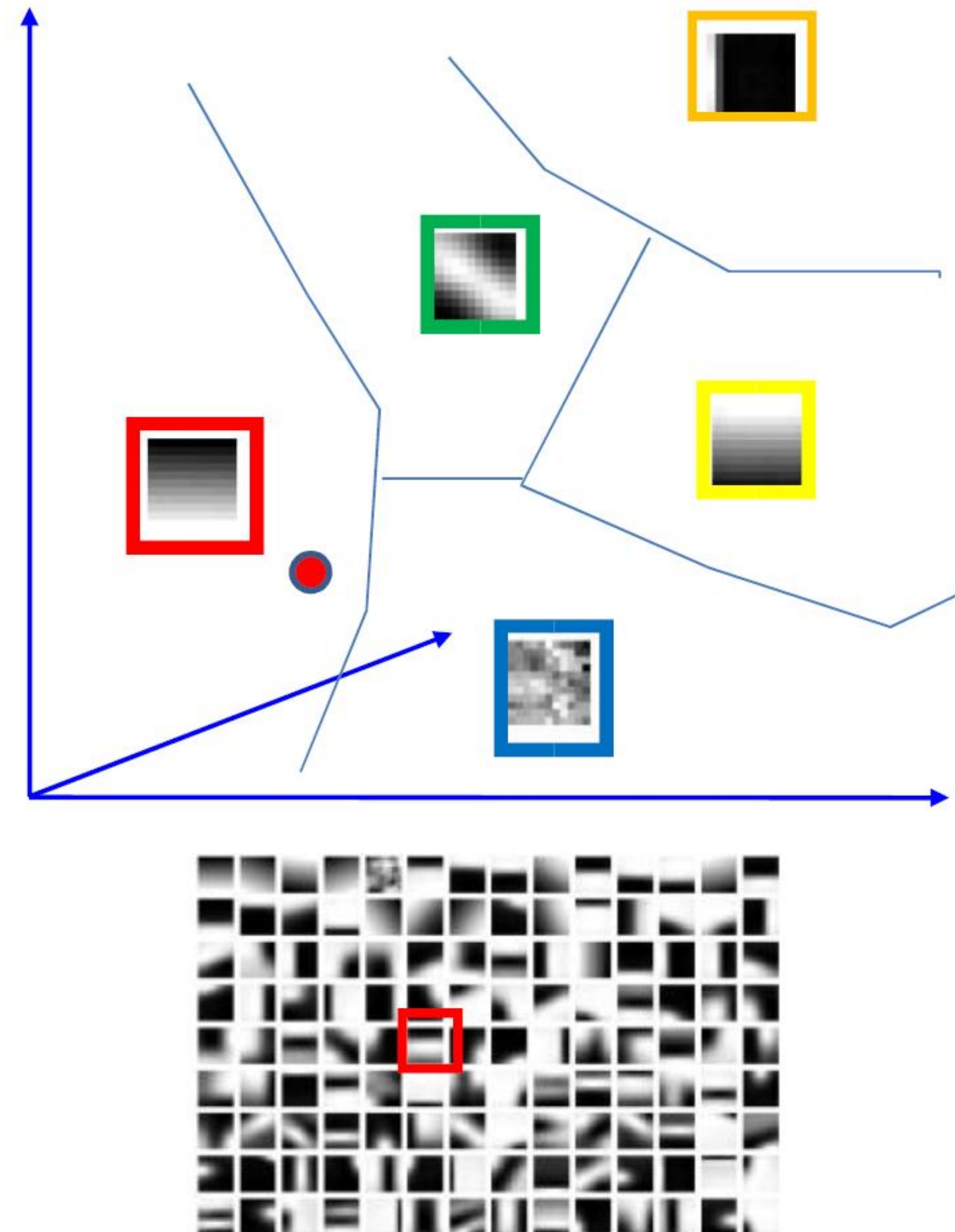


二、Bag of Words

3. Bag of Word representation



多类支撑向量机训练分类器

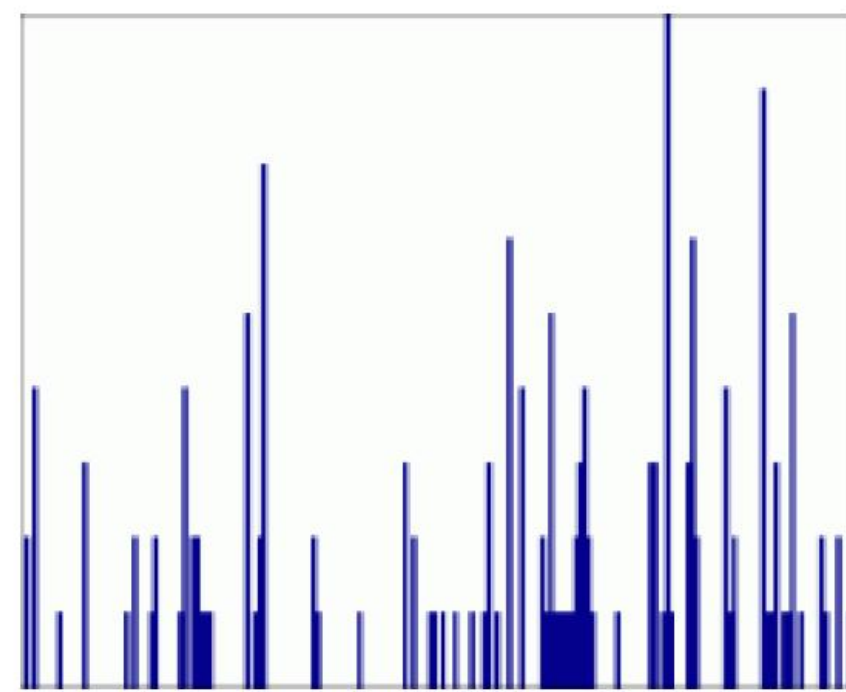
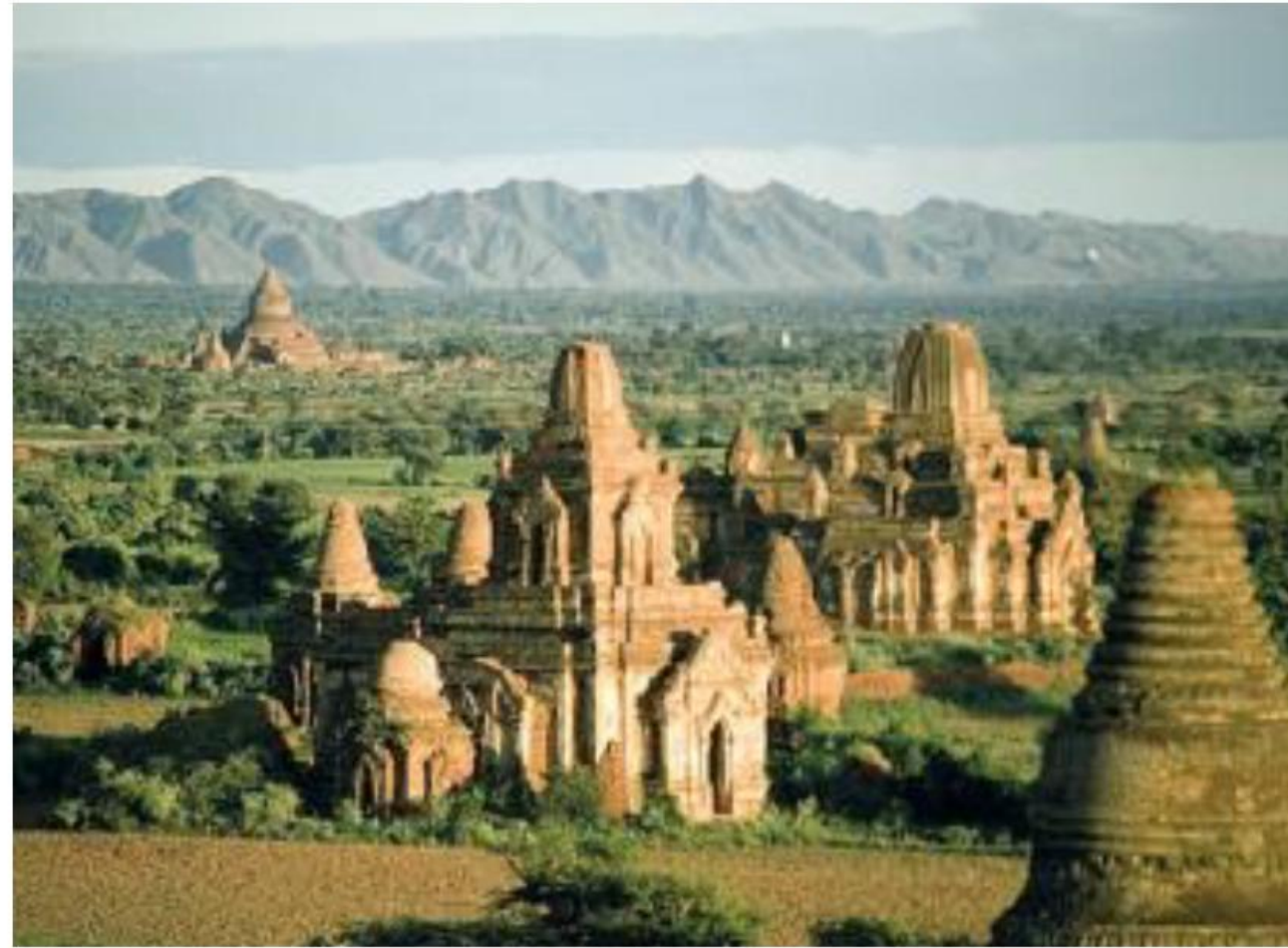


二、Bag of Words

Spatial pyramid representation

Extension of a bag of features

Locally orderless representation at several levels of resolution



level 0

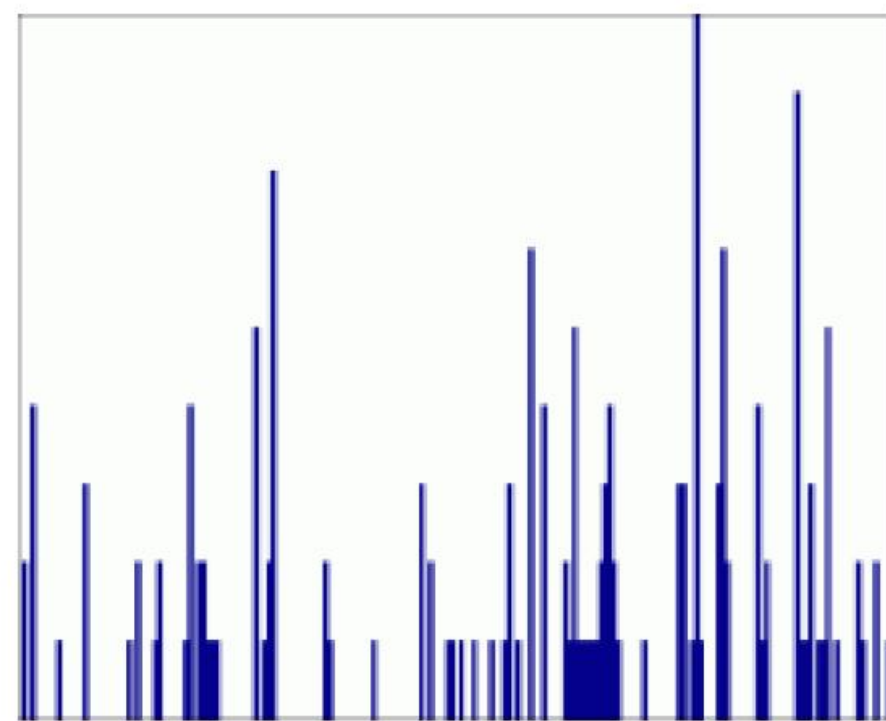
全局信息看起来一样，其实细节不一样，需要考虑空间关系

二、Bag of Words

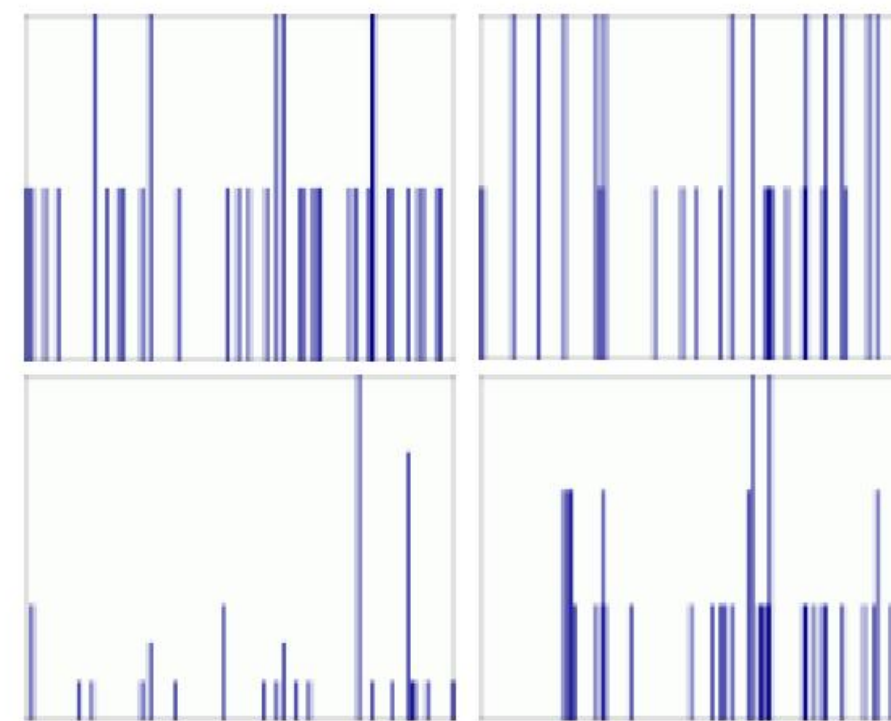
Spatial pyramid representation

Extension of a bag of features

Locally orderless representation at several levels of resolution



level 0



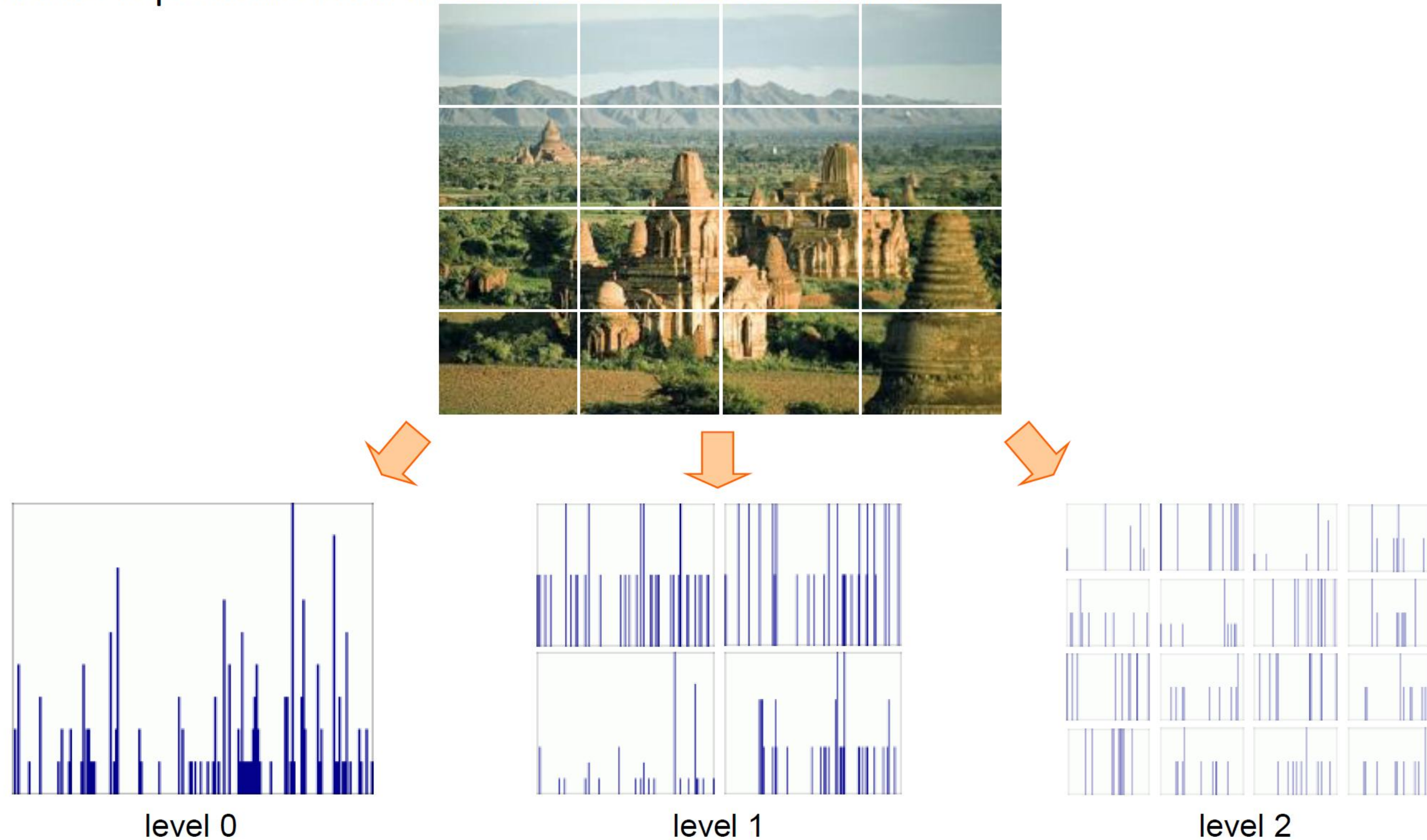
level 1

二、Bag of Words

Spatial pyramid representation

Extension of a bag of features

Locally orderless representation at several levels of resolution



某视觉词典共有 5 个视觉单词，其聚类中心分别为

$$w_1 = 1, \quad w_2 = 3, \quad w_3 = 5, \quad w_4 = 8, \quad w_5 = 10$$

现有两幅图像 A 和 B ，它们提取出的局部特征描述子（一维，便于计算）如下：

$$A : \{1.2, 2.7, 4.6, 5.4, 7.6, 8.2, 9.7, 5.1\}$$

$$B : \{0.9, 3.2, 3.8, 4.9, 8.7, 9.4, 9.8, 10.0\}$$

设每个局部特征都分配给**距离最近**的视觉单词（硬分配，采用欧氏距离）。

请完成下列问题：

1. 分别求图像 A 和 B 中每个局部特征对应的视觉单词编号；
2. 写出图像 A 和 B 的 BoW 频数直方图；
3. 对两个直方图分别做 **L1 归一化**；
4. 计算两个归一化后向量的**欧氏距离**；